llmXive Automated Review of AgentDoG 1.5: A Lightweight and Scalable Alignment Framework for AI Agent Safety and Security

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (8 lenses): Claim Accuracy, Figure Critic, Jargon Police, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The manuscript makes several high-impact factual claims that are not fully supported by the provided text or citations.

First, the central claim that AgentDoG achieves performance “comparable to GPT-5.4” and “Gemini-3.1-Pro” (Abstract, Introduction) relies entirely on citations to sources with future dates (e.g., `openai_gpt54_2026`, `anthropic_claude_opus_46_2026`, `gemini3`). As a third-party preprint reviewer, I cannot verify the existence or content of these unreleased system cards. If these sources do not contain the specific benchmark results cited, the claim of parity with frontier models is unsupported. The paper must either provide the actual data from these sources or qualify the claim as a projection based on available data.

Second, there is a significant internal inconsistency in the reported results. Table 1 in the Introduction (e006) reports AgentDoG-4B achieving 92.8% F1 on ATBench, while Table 2 in Section 4 (e000) reports 72.4% F1 on the same benchmark for the same model. The text does not explain this discrepancy (e.g., different evaluation settings, model variants, or data splits). This contradiction undermines the accuracy of the performance claims.

Third, the claim of training on “around 1k samples” (Abstract, Sec 4.2) is derived from a purification process starting with 5,973 tools, but the exact number of final training trajectories is not explicitly stated. While “around 1k” is a reasonable approximation, the lack of a precise figure in the methodology or tables makes the claim difficult to verify against the “influence-function purification” description.

Finally, the claim of reducing Docker overhead by “two orders of magnitude” (Abstract) and to “1/100” (Intro) is supported by a peak memory figure of <2.5GB (Sec 5.2.2) but lacks a direct baseline comparison of Docker memory usage for the *same* tasks. Without a stated baseline (e.g., “Docker environments consume ~250GB for equivalent workloads”), the “100x” reduction is an unsupported extrapolation.

Action items.

- [**science**] The claim that AgentDoG matches GPT-5.4 performance (Abstract, Intro) relies on citations to unreleased/future-dated sources (e.g., openai_gpt54_2026, anthropic_claude_opus_46_2026). Verify these citations exist and that the comparison data is not hallucinated or based on speculative system cards.
- [**writing**] The claim of ‘around 1k samples’ for training (Abstract, Sec 4.2) is supported by a purification step from 5,973 tools, but the exact final count of training trajectories used for the 0.8B-8B models is not explicitly stated in the text or tables, making the ‘1k’ figure an unverified approximation.
- [**science**] Table 1 (e006) and Table 2 (e000) present conflicting results for AgentDoG-4B on R-Judge (92.2% vs 91.8% F1) and ATBench (72.4% vs 92.8% F1). The text does not explain which table represents the final results or why the metrics differ so drastically between the two tables.
- [**science**] The claim that the environment reduces Docker overhead by ‘two orders of

magnitude' (Abstract) and to '1/100' (Intro) is supported by a memory claim (<2.5GB) but lacks a direct comparison of Docker memory usage for equivalent tasks, making the specific '100x' ratio an unsupported extrapolation.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure effectively communicates the performance of AgentDoG 1.5 against baselines with clear numerical labels on bars. However, the orange text color for the x-axis labels matches the bar color, slightly reducing readability, and the steep rotation of the labels is a minor aesthetic issue.

Figure 2

Figure 2 clearly illustrates the trajectory-level safety evaluation process by contrasting the task definitions and outputs for safe versus unsafe agent behaviors. The visual layout effectively distinguishes between the input trajectory and the resulting classification, supporting the caption's description.

Figure 3

The figure provides a clear visual overview of the data engine and training pipeline. However, the caption contains significant grammatical errors where the model name is missing (e.g., 'Pipeline of .'), and the specific statistics listed in the 'Risk Sampling' box are not visually represented in the diagram.

Figure 4

The figure illustrates two scenarios but fails to visually distinguish between 'benign data' and 'deceptive instructions' in the left panel as claimed in the caption. Additionally, the red highlighting lacks a legend to explain its specific meaning regarding attribution ranking.

Figure 5

The figure displays a trajectory comparison with scores but fails to visually represent the 'attribution analysis' described in the caption, creating a disconnect between the text and the visual data. Additionally, the figure title lacks the specific context found in the caption.

Figure 6

The figure presents a clear visual taxonomy of the dataset, but the caption contains grammatical errors with missing text, and the charts lack specific data labels to support the distribution analysis.

Figure 7

The figure provides a clear visual workflow for the dual-scenario synthesis pipeline, but the caption is grammatically incomplete. Additionally, the logic of the ‘malicious query attack’ path appears counter-intuitive by pairing a malicious request with a benign context to generate a safety task.

Figure 8

The figure effectively visualizes scalability metrics, but the caption’s claim that memory footprint remains ‘highly stable’ is contradicted by the visible upward trend in the Peak RSS data across all subplots.

Figure 9

The figure provides a clear visual schematic of the system architecture, but the caption is insufficiently descriptive, failing to explain the specific modules or the interaction flow depicted in the diagram.

Figure 10

The figure provides a clear visual overview of the two application modes (Training-Based and Training-Free), but the caption is too generic, failing to identify the framework by name or explain the specific workflow steps depicted in the diagram.

Figure 11

Figure 11 is a clear and well-structured taxonomy diagram that effectively visualizes the three-dimensional agentic safety framework. The legend is comprehensive, explicitly defining the color and border styles used to distinguish between OpenClaw and Codex customizations and strengthened items, which aligns perfectly with the caption’s description.

Figure 12

Figure 12 is a clear and well-structured diagram that effectively illustrates the ATBench family hierarchy. The visual layout logically connects the shared taxonomy and diagnosis task to the specific customizations of ATBench, ATBench-Claw, and ATBench-Codex, fully supporting the claims in the caption.

Action items.

- [**writing**] Figure 1: The x-axis labels for the 'AgentDoG' model variants are colored orange, while the bars are also orange, creating a visual redundancy that makes the text harder to distinguish from the bars compared to the grey bars.
- [**writing**] Figure 1: The x-axis labels are rotated at a steep angle, which is necessary for fit but makes the model names (e.g., 'Qwen3-235B') difficult to read quickly.
- [**writing**] Figure 3: The caption repeatedly refers to 'Building Pipeline of .' and '...of .' with a missing model name, likely due to a placeholder variable not being replaced in the final text.
- [**writing**] Figure 3: The 'Risk Sampling' box lists '15 Risk Sources', '21 Failure Modes', and '11 Real-world Harms', but the figure does not visually depict these specific counts or categories, making the text feel disconnected from the visual flow.
- [**science**] Figure 4: The caption claims the left panel isolates 'embedded deceptive instructions,' but the visual highlights the 'Environment' block (containing resume data) and the 'Key Impact Step' (containing the instruction), failing to visually distinguish the 'benign data' from the 'deceptive instruction' as promised.
- [**writing**] Figure 4: The caption states 'Highlighted regions indicate the top-ranked components,' yet the figure uses red text highlighting without a legend or key to define what the red color signifies (e.g., attribution score, confidence, or specific token importance).
- [**science**] Figure 5: The caption claims the figure shows 'Comparative attribution analysis' and highlights 'high-impact drivers' (Steps 2, 3, 4, 5), but the rendered image displays a chronological 'Trajectory' of actions and thoughts with 'Traj Scores' rather than an attribution heatmap or importance map. The visual content does not match the caption's description of an attribution analysis.
- [**writing**] Figure 5: The figure title 'Attribution Comparison: AgentDoG vs. Base Model' is generic and does not specify the scenario (WeChat red packet) or the specific models (Qwen3-4B) mentioned in the caption, making the figure standalone context poor.
- [**writing**] Figure 6: The caption contains a grammatical error and missing subject: 'filtered agentic safety SFT data by .' and 'selected by ,' (likely missing 'AgentDoG').
- [**science**] Figure 6: The donut charts lack numerical labels or percentages on the segments, making it impossible to verify the distribution claims or compare segment sizes accurately.
- [**writing**] Figure 7: The caption 'The dual-scenario environment synthesis pipeline for agentic safety RL' is grammatically incomplete and lacks the specific subject name (e.g., 'AgentDoG') that is present in other captions (e.g., Figure 3, Figure 6).
- [**science**] Figure 7: The 'malicious query attack' path depicts a 'Malicious User Request' combined with a 'Benign Environment Context' to create a 'Safety Task Bundle'. This contradicts standard safety evaluation logic where a malicious request is typically paired with a vulnerable or corrupted context to test for failure, whereas the 'Benign User Request'

path pairs with a 'Corrupted Environment Context' to create a 'Clean Task Bundle', which seems counter-intuitive for a safety synthesis pipeline.

- [**science**] Figure 8: The caption claims memory footprint remains 'highly stable' and consumes 'less than 2.5 GB', but the 'Peak RSS' data (orange dashed lines) in all three subplots shows a clear, monotonic increase with workload, contradicting the claim of stability.
- [**writing**] Figure 8: The y-axis label 'Per-Env Time (ms)' is rotated 90 degrees and difficult to read; it should be horizontal or the font size increased for clarity.
- [**writing**] Figure 9: The caption 'Our online agent safety guardrail pipeline' is too generic and fails to describe the specific components shown (e.g., AgentDoG Guardrail, Trajectory Formatter, Agent Runtime) or the data flow between the 'Autonomous Agent Execution' and 'Online Guardrail' sections.
- [**writing**] Figure 9: The text 'Judges full trajectory' inside the AgentDoG Guardrail box is grammatically awkward and likely a typo for 'Judges the full trajectory' or 'Judges full trajectory context'.
- [**writing**] Figure 10: The caption describes the figure as 'A lightweight and scalable alignment framework' but fails to name the framework (AgentDoG) or explain the specific components shown (Application 1 vs. Application 2), making the diagram difficult to interpret without reading the main text.
- [**writing**] Figure 10: The central logo contains the text 'AgentDoG', but this name is not explicitly defined in the caption, which refers to the system only as 'A lightweight and scalable alignment framework'.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific jargon and undefined acronyms, which creates a barrier for readers outside the immediate sub-field of AI agent safety. While the technical contributions are significant, the presentation frequently assumes prior knowledge that should be explicitly stated.

In the Introduction and Section 2, acronyms such as "MCP," "SFT," and "RL" are used repeatedly without being defined at their first occurrence. This forces the reader to guess or search for definitions, disrupting the flow of reading. Similarly, "GDPO" is introduced in Section 3.3 without its full expansion, despite being a core component of the proposed method.

The term "trajectory" is used extensively to describe agent execution paths (e.g., Section 3.1, Section 5). While standard in the field, a brief plain-language clarification (e.g., "a sequence of agent actions and observations") would make the text more accessible to a broader audience.

The benchmark names "ATBench," "ATBench-Claw," and "ATBench-Codex" are introduced with technical precision but lack a simple explanation of their relationship and purpose for the general reader.

Furthermore, terms like "finite-state" (Section 3.4) and metrics like "ASR" and "TTFT" (Section 5.3) are presented without sufficient context. Replacing "finite-state" with "simplified deterministic" and explicitly defining these metrics upon first use would significantly improve clarity. The authors should review the entire text to ensure every acronym is defined at first use and that specialized terminology is either explained or replaced with plainer alternatives where possible.

Action items.

- [**writing**] Define 'MCP' (Model Context Protocol) at first use in Section 1 or 2. Currently used without definition.
- [**writing**] Define 'SFT' (Supervised Fine-Tuning) and 'RL' (Reinforcement Learning) at first use. These acronyms appear frequently without prior definition.
- [**writing**] Define 'GDPO' (Group Reward-Decoupled Normalization Policy Optimization) at first use in Section 3.3. The acronym is used before the full term is introduced.
- [**writing**] Replace 'trajectory' with 'sequence of actions' or 'execution history' in the Introduction and Section 3.1 to improve accessibility for non-specialists.
- [**writing**] Define 'ATBench' and its variants (ATBench-Claw, ATBench-Codex) clearly at first mention in Section 2. The distinction between the base and customized versions needs a plain-language explanation.
- [**writing**] Replace 'finite-state' with 'simplified deterministic' when describing the simulation environment in Section 3.4 to reduce jargon density.
- [**writing**] Define 'ASR' (Attack Success Rate) and 'TTFT' (Time To First Token) in Section 5.3 where they are first used in the table context.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The manuscript exhibits significant overreach in its claims regarding performance equivalence, resource efficiency, and data efficiency, which are not fully substantiated by the provided experimental data.

First, the Abstract and Introduction repeatedly claim that AgentDoG achieves performance "comparable to GPT-5.4" and "matches closed-source models." However, the results tables (Table 1, Table 2) do not present a direct, head-to-head comparison with GPT-5.4 on the specific

ATBench or R-Judge metrics where this equivalence is asserted. While GPT-5.4 is listed in Table 1, the specific numbers for the "matching" claim are ambiguous or missing in the context of the specific benchmarks discussed in the text. Claiming parity with a frontier model without a clear, side-by-side statistical comparison in the results section is an over-claim.

Second, the claim of reducing "Docker overhead by two orders of magnitude" (Abstract, Section 1) is a strong quantitative assertion that lacks the necessary baseline data. The paper mentions a peak memory of <2.5 GB for the synthesized environments (Section 5.1.2), but it fails to provide the corresponding memory or latency figures for the Docker-based baselines (e.g., SWE-Bench, AgentHazard) used for comparison. Without the baseline numbers, the "1/100" or "two orders of magnitude" claim is unsupported and potentially misleading.

Third, the assertion of training with "only around 1k samples" (Abstract, Section 4.2) requires clarification to avoid over-interpretation. The text describes generating 5,973 unique tools and 32,787 pairs, then using influence functions to select a subset. The paper must explicitly state whether the final training set is exactly ~1k samples and provide a justification for why such a small dataset yields state-of-the-art results compared to models trained on significantly larger corpora. Without this, the claim risks overstating the data efficiency of the method.

Finally, the text states that AgentDoG-4B "outperforms Qwen3.5-397B" (Section 4.4.1) while also claiming to "match" GPT-5.4. The provided Table 1 shows AgentDoG-4B at 72.4% Acc and Qwen3.5-397B at 66.8% Acc. While the numbers show an improvement, the phrasing "outperforms" a 397B model while "matching" a frontier model (GPT-5.4) creates a logical tension regarding the magnitude of the improvement. The paper should temper these claims to reflect the specific, modest margins observed in the data rather than implying a broad superiority over massive models.

Action items.

- [**science**] The manuscript exhibits significant overreach in its claims regarding performance equivalence, resource efficiency, and data efficiency, which are not fully substantiated by the provided experimental data. First, the Abstract and Introduction repeatedly claim that AgentDoG achieves performance "comparable to GPT-5.4" and "matches closed-source models." However, the results tables (Table 1, Table 2) do not present a direct, head-to-head comparison with GPT-5.4 on the specific ATBench or R-Judge m

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a valuable framework for AI agent safety, but several ethical and safety concerns require attention before publication.

First, the dual-use risk of the synthetic data is significant. The appendices (specifically Appendix C) provide detailed prompts for generating adversarial trajectories, including specific examples of harmful actions like unauthorized WhatsApp messaging and authorization bypass. While intended for defensive training, releasing such granular attack patterns without explicit ethical constraints could facilitate malicious use. The authors must include a clear statement in the Ethics or Data Availability section restricting the dataset's use to safety research, specifying a restrictive license, and considering the redaction of realistic exploit payloads (e.g., specific phone numbers) in the public release.

Second, the evaluation of the online guardrail (Section 5) involves real-time intervention. The manuscript does not explicitly confirm that these tests were conducted in a fully isolated sandbox environment. Given the risk of false positives blocking legitimate actions, a brief statement confirming that all guardrail evaluations were performed in non-production, controlled environments is necessary to ensure ethical deployment practices.

Finally, the methodology for selecting the training subset via influence functions (Section 4.2) lacks discussion on sample diversity. There is a risk that the selection process could inadvertently filter out rare or complex safety scenarios, leading to a model blind to edge cases. A sentence clarifying how the authors ensured the diversity of the selected safety samples would strengthen the ethical rigor of the work.

Action items.

- [**writing**] The paper details synthetic attack generation (e.g., WhatsApp injection) in Appendix C. Authors must add an ethics statement restricting dataset use to safety research, consider redacting specific exploit payloads to prevent dual-use, and clarify the license.
- [**writing**] The online guardrail evaluation (Section 5) lacks confirmation that testing occurred in isolated, non-production environments. A statement confirming sandbox-only deployment is required to address potential service disruption risks.
- [**writing**] The influence-function purification method (Section 4.2) does not address potential bias in sample selection. Authors should clarify how diversity of safety risks was maintained to avoid filtering out edge cases.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The manuscript presents a compelling framework for agent safety, but the scientific evidence supporting the central claims requires significant strengthening. The most critical issue is the lack of statistical rigor in the experimental results. The claim that AgentDoG-4B matches or exceeds GPT-5.4 (Table 1, Sec 4.4) is presented with point estimates only. Given the relatively small test sets (e.g., ATBench-Claw has 500 trajectories), the absence of confidence intervals or p-values makes it impossible to assess whether the reported improvements are statistically significant or merely artifacts of variance. The authors must re-run evaluations with multiple seeds or bootstrap resampling to provide 95% confidence intervals for all key metrics.

Furthermore, the data efficiency claim of training on "around 1k samples" (Abstract, Sec 4.2) is not sufficiently substantiated. The paper describes an influence-function purification pipeline but does not disclose the exact number of samples retained, the specific influence function thresholds used, or a sensitivity analysis. Without showing performance curves across varying dataset sizes (e.g., 500, 1000, 2000 samples), the claim of high efficiency is vulnerable to the alternative explanation that the model simply overfits a small, highly curated subset. A learning curve analysis is essential to validate the "lightweight" assertion.

Finally, the scalability claims regarding the finite-state environment (Sec 5.1.2) lack a direct empirical comparison. While the authors state memory usage is <2.5GB, they do not provide the corresponding metrics for the Docker-based baselines (e.g., SWE-Bench) under the same concurrency load. To support the "two orders of magnitude" reduction claim, a side-by-side resource profiling table is required. Without these controls, the efficiency argument remains anecdotal.

Action items.

- [**science**] The manuscript presents a compelling framework for agent safety, but the scientific evidence supporting the central claims requires significant strengthening. The most critical issue is the lack of statistical rigor in the experimental results. The claim that AgentDoG-4B matches or exceeds GPT-5.4 (Table 1, Sec 4.4) is presented with point estimates only. Given the relatively small test sets (e.g., ATBench-Claw has 500 trajectories), the absence of confidence intervals or p-values makes it impos

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical rigor of the evaluation section requires significant strengthening before the claims can be accepted.

First, the central claim of achieving state-of-the-art performance with "around 1k samples" (Abstract; Section 4.2) is statistically unsupported. The manuscript does not specify the exact number of samples used, nor does it provide a variance analysis. Given the high dimensionality of the task, a sample size of ~1,000 is small. The authors must report the exact count, the standard deviation of performance across multiple random seeds of the training subset, and ideally, a learning curve or ablation study demonstrating that performance plateaus at this sample size. Without this, the "lightweight" claim is anecdotal.

Second, the reporting of results in Tables 1, 2, and 3 (Sections 4.4.1, 4.4.2, 5.3) relies exclusively on point estimates (Accuracy, F1, ASR) without confidence intervals (CIs) or standard deviations. This is particularly problematic for benchmarks with small test sets, such as CIK-Bench (n=35 trajectories, Table 3, e006). A difference in ASR from 94.29% to 42.86% on 35 samples is a large effect size but requires a formal statistical test (e.g., McNemar's test or Fisher's exact test) to establish significance. Reporting p-values or 95% CIs for all comparative metrics is mandatory to distinguish signal from noise.

Third, the data purification strategy using influence functions (Section 4.2, Eq 1) is presented as a deterministic filter. The authors should provide statistical evidence that the selected subset is indeed "high-value." This could be done by comparing the gradient norms or loss distributions of the selected vs. discarded samples, or by showing that random sampling of the same size yields significantly lower performance. Currently, the selection process is a "black box" with no statistical validation of its efficacy.

Finally, the RL results in Section 5.2 (Table 4, e006) compare SFT, RL, and SFT+RL. The improvements are presented as absolute percentage gains. Given the stochastic nature of RL training, these results should be averaged over multiple seeds with reported standard deviations to ensure the observed gains are reproducible and not due to random initialization variance.

Action items.

- [**science**] The statistical rigor of the evaluation section requires significant strengthening before the claims can be accepted. First, the central claim of achieving state-of-the-art performance with "around 1k samples" (Abstract; Section 4.2) is statistically unsupported. The manuscript does not specify the exact number of samples used, nor does it provide a variance analysis. Given the high dimensionality of the task, a sample size of ~1,000 is small. The authors must report the exact count, the standar

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a comprehensive framework for AI agent safety, but the writing quality requires minor revisions to ensure precision and flow. The most significant issue is the inconsistent use of numerical approximations. In Section 3.2 (Data Preparation), the authors repeatedly state that the model is trained on "around 1k samples." Given that Section 2.3 explicitly defines the ATBench base dataset as containing "1,000 trajectories," the use of "around" creates ambiguity. The authors should replace "around 1k" with the exact integer count (e.g., "1,024" or "1,000") to align with the precision expected in scientific reporting.

Additionally, there are several sentence fragments that disrupt the reading flow. In Section 4.1, the text reads: "Mixture: 1:2 safety-critical to benign (50,000 benign trajectories from ToolBench, ToolAlpaca, ToolACE)." This is not a complete sentence and should be integrated into the preceding or following sentence to form a grammatically complete thought. Similarly, in Section 5.2, the benchmark "CIK-Bench" is introduced without a prior definition or full name expansion, which may confuse readers unfamiliar with the specific acronym.

Finally, the abstract claims the framework updates the taxonomy for "Codex/OpenClaw scenarios," while the body of the paper (Sections 2.3 and 3.1) treats ATBench-Claw and ATBench-Codex as distinct benchmark instances with specific customizations. The abstract should be refined to clarify whether the taxonomy update is a unified framework applied to both or if it involves distinct updates for each scenario, ensuring consistency with the detailed descriptions in the main text. Addressing these points will significantly improve the clarity and professional polish of the manuscript.

Action items.

- [**writing**] In Section 3.2 (Data Preparation), the phrase 'around 1k samples' is used repeatedly. Replace with a precise integer (e.g., '1,024 samples') or a specific range to maintain scientific rigor and consistency with the '1,000 trajectories' mentioned in Section 2.3.
- [**writing**] Section 4.1 contains a sentence fragment: 'Mixture: 1:2 safety-critical to benign (50,000 benign trajectories from ToolBench, ToolAlpaca, ToolACE).' Rewrite as a complete sentence, e.g., 'The final dataset maintains a 1:2 mixture of safety-critical to benign trajectories, incorporating 50,000 benign samples from...'
- [**writing**] In Section 5.2, the text states '35 retained CIK-Bench cases' but the table caption refers to 'CIK-Bench' without defining the acronym earlier in the section. Ensure 'CIK-Bench' is defined upon first use or replaced with the full name if it is not a standard benchmark.
- [**writing**] The abstract claims the framework updates the safety taxonomy for 'Codex/OpenClaw scenarios,' but the Introduction (Section 1) and Section 2.3 distinguish between 'ATBench-Claw' and 'ATBench-Codex' as separate benchmark instances. Clarify in the abstract whether the taxonomy update applies to both distinct scenarios or a unified one.